Assignment 4 – Task 1

Sequential Handwritten Kannada Digit Classification using Recurrent Neural Networks

Course: LLM **Task:** Row-by-row sequential modeling

Summary

This task formulates each 28 × 28 grayscale Kannada numeral image as a sequence of 28 row vectors and compares six recurrent classifiers built from vanilla RNN, LSTM, and GRU cells. The experiments vary hidden size (64 or 128) and recurrent depth (one or two layers). Model selection is based on validation accuracy, while Dig-MNIST provides an additional labeled evaluation set.

Selected criterion	Model	Training accuracy	Validation accuracy	Dig-MNIST accuracy
Best validation	LSTM_H128_L2	99.72%	99.30%	74.26%
Best Dig-MNIST score	GRU_H128_L2	99.74%	99.17%	74.36%

Key observation. LSTM_H128_L2 achieved the highest validation accuracy (99.30%), whereas GRU_H128_L2 achieved the highest Dig-MNIST accuracy (74.36%). This distinction is important: the assignment guide specifies validation accuracy for selecting the model, rather than choosing the model solely by the held-out Dig-MNIST result.

1. Overview and Objectives

The objective is to study how recurrent architectures process a static image when it is represented sequentially. The experiment covers all ten Kannada digit classes (0–9), compares recurrent cell types and model sizes, tracks average training cross-entropy loss, and evaluates classification performance using accuracy and a confusion matrix.

The report follows the layout and concise, figure-led presentation of the Assignment 3 reference report: summary, objectives, methodology, experiment results, visual analysis, inferences, and conclusion.

2. Dataset Representation and Preprocessing

Each image contains 784 grayscale pixel values. The values are reshaped to a 28 × 28 matrix, normalized from [0, 255] to [0, 1], and passed to the recurrent model as 28 time steps with 28 features per step. Thus, the input tensor has shape (number of images, 28, 28). Each time step corresponds to one image row.

The supplied workflow uses train.csv to form a stratified training/validation split (90%/10%) and uses the labeled Dig-MNIST.csv data for additional evaluation. The unlabeled test(1).csv file can be used for predictions, but accuracy cannot be computed without ground-truth labels.

3. Methodology and Model Architectures

The recurrent layer reads image rows in order and updates its hidden representation at each time step. The final time-step output is passed to a linear classifier that produces ten class scores. Cross-entropy loss is used for optimization, and the Adam optimizer updates model parameters through backpropagation.

Model	Cell	Hidden units	Layers	Epochs	Final training loss
RNN_H64_L1	RNN	64	1	54	0.034786
RNN_H128_L2	RNN	128	2	100	0.021368
LSTM_H64_L1	LSTM	64	1	36	0.009835
LSTM_H128_L2	LSTM	128	2	31	0.008071
GRU_H64_L1	GRU	64	1	40	0.011639
GRU_H128_L2	GRU	128	2	21	0.007419

The six configurations provide a controlled comparison of cell type, hidden size, and depth. Vanilla RNN uses a simpler recurrent update. LSTM and GRU use gating mechanisms to regulate information flow through the sequence, which can help preserve useful information across time steps.

3.1 Convergence rule

Training follows the assignment criterion: stop when the absolute difference between average training loss in consecutive epochs is below 10^-4, subject to a maximum of 100 epochs. A small loss difference indicates that the training loss has stabilized; it does not, by itself, guarantee high validation accuracy or good generalization. RNN_H128_L2 reached the 100-epoch cap, so it should be reported as capped rather than assumed to have met the convergence threshold.

4. Model Comparison

Model	Train acc.	Validation acc.	Dig-MNIST acc.	Final loss	Epochs
RNN_H64_L1	99.08%	98.55%	65.51%	0.034786	54
RNN_H128_L2	99.20%	98.72%	71.62%	0.021368	100
LSTM_H64_L1	99.72%	99.10%	72.35%	0.009835	36
LSTM_H128_L2	99.72%	99.30%	74.26%	0.008071	31
GRU_H64_L1	99.80%	98.97%	71.95%	0.011639	40
GRU_H128_L2	99.74%	99.17%	74.36%	0.007419	21

LSTM_H128_L2 provides the strongest validation result at 99.30%, with 99.72% training accuracy and 74.26% Dig-MNIST accuracy. GRU_H128_L2 has the best Dig-MNIST accuracy (74.36%) and a validation accuracy of 99.17%. The difference in Dig-MNIST performance is small, while the validation criterion favors the two-layer 128-unit LSTM.

5. Training Error and Convergence

Figure 1 compares the average training cross-entropy loss over epochs for all six configurations. All curves fall sharply during the early epochs, indicating rapid learning of useful class-discriminative patterns. The curves then flatten at relatively low loss values. The exact number of epochs differs by configuration because the stopping condition is based on the change in training loss, with 100 epochs as a safety cap.

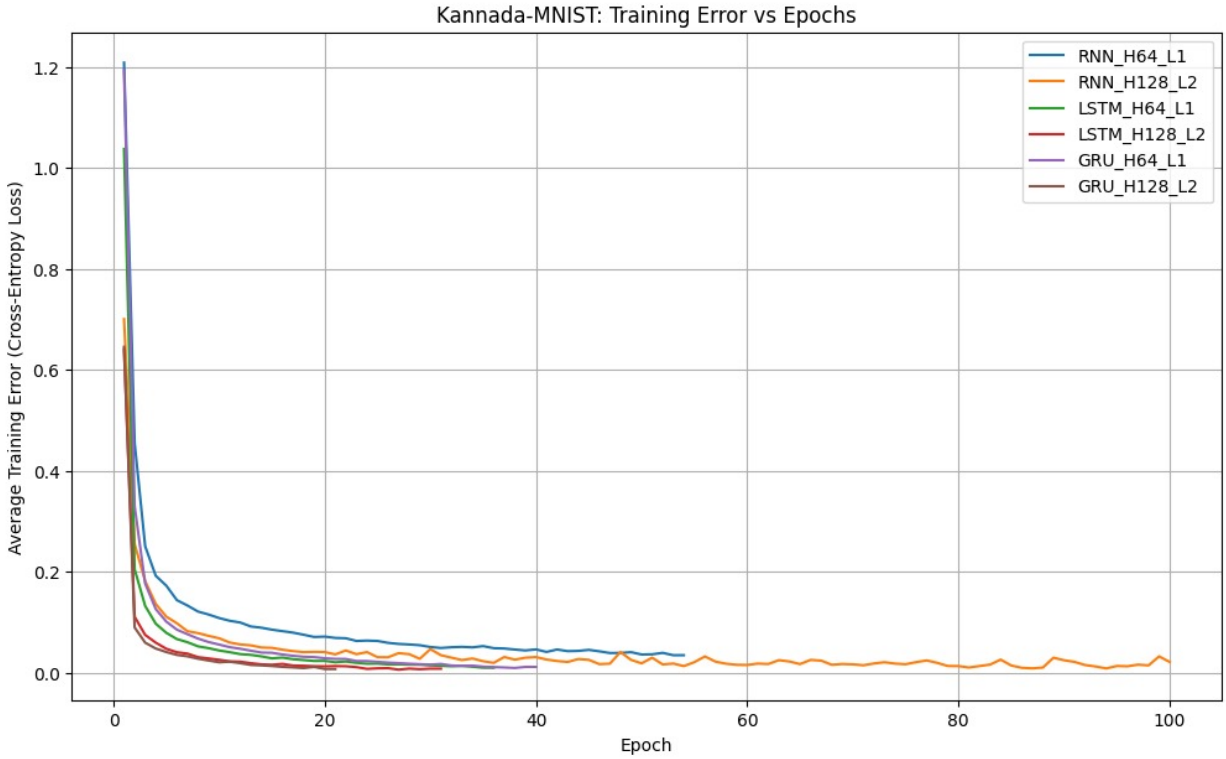


Figure 1. Average training error versus epochs for the six RNN, LSTM, and GRU configurations (supplied experimental plot).

The single-layer LSTM and GRU models reach low loss in fewer epochs than the two-layer RNN configuration. The RNN_H128_L2 curve continues to the 100-epoch limit and exhibits small fluctuations late in training. Mini-batch optimization can cause such variation. A lower training loss is useful evidence of fit, but model selection should also consider validation performance and held-out evaluation.

5.1 Accuracy comparison

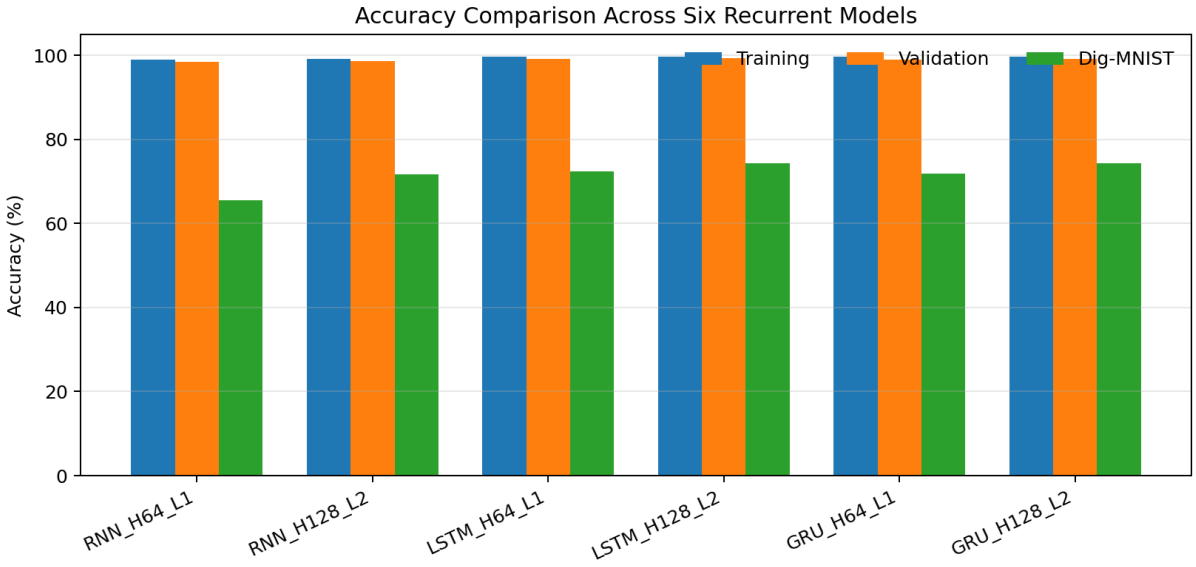


Figure 2. Training, validation, and Dig-MNIST accuracy for each tested configuration, computed from the supplied results workbook.

6. Best Model and Confusion Matrix

The highest validation accuracy belongs to LSTM_H128_L2 (99.30%), so it is selected under the assignment's validation-based model-selection rule. Its Dig-MNIST accuracy is 74.26%, substantially lower than its training and validation scores. This gap suggests that the additional evaluation set is more challenging or differs in distribution from the train.csv split; the result should not be interpreted as proof of a specific cause without further investigation.

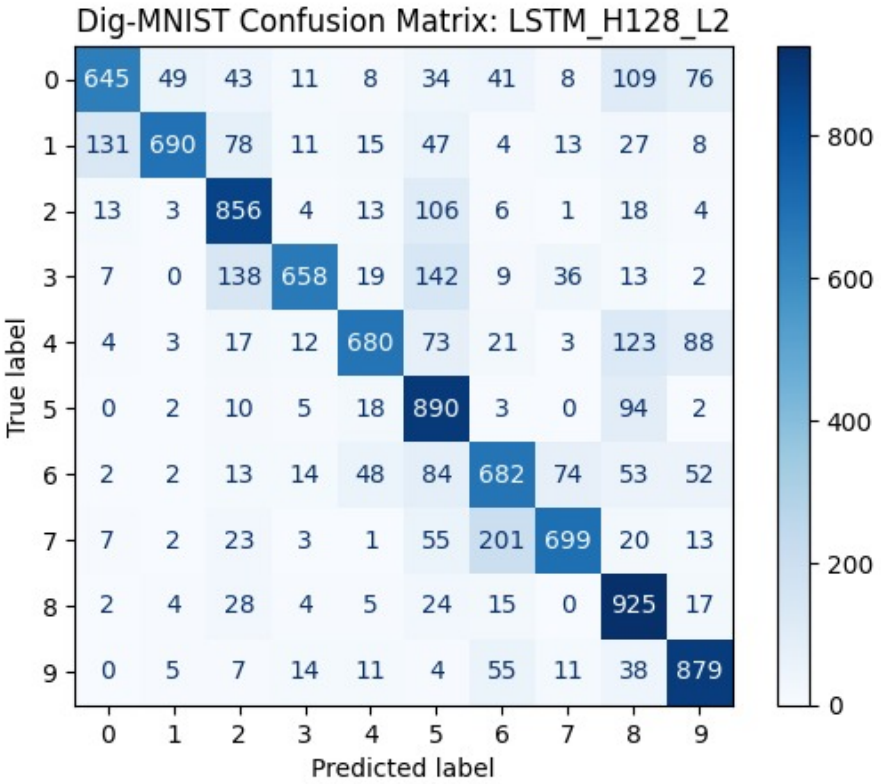


Figure 3. Dig-MNIST confusion matrix for LSTM_H128_L2. Rows represent true labels and columns represent predicted labels.

The diagonal entries represent correctly classified samples, while off-diagonal entries indicate errors. The matrix shows 645 correct predictions for class 0, 690 for class 1, 856 for class 2, 658 for class 3, 680 for class 4, 890 for class 5, 682 for class 6, 699 for class 7, 925 for class 8, and 879 for class 9. The strongest visible confusions include true class 3 predicted as class 5 (142 cases), true class 1 predicted as class 0 (131), true class 4 predicted as class 8 (123), and true class 0 predicted as class 8 (109).

These errors indicate that certain handwritten forms are visually similar to the model, or that their appearance varies across examples. Class-wise inspection and additional validation on data resembling Dig-MNIST would help investigate the source of these confusions.

7. Qualitative Visualization of Kannada Digits

Figure 4 shows five example images for each of the ten digit classes. Each small image is a 28 × 28 grayscale sample displayed on a black background. The grid helps verify that the dataset contains diverse handwriting styles and that the digit classes have recognizable but sometimes overlapping visual patterns.

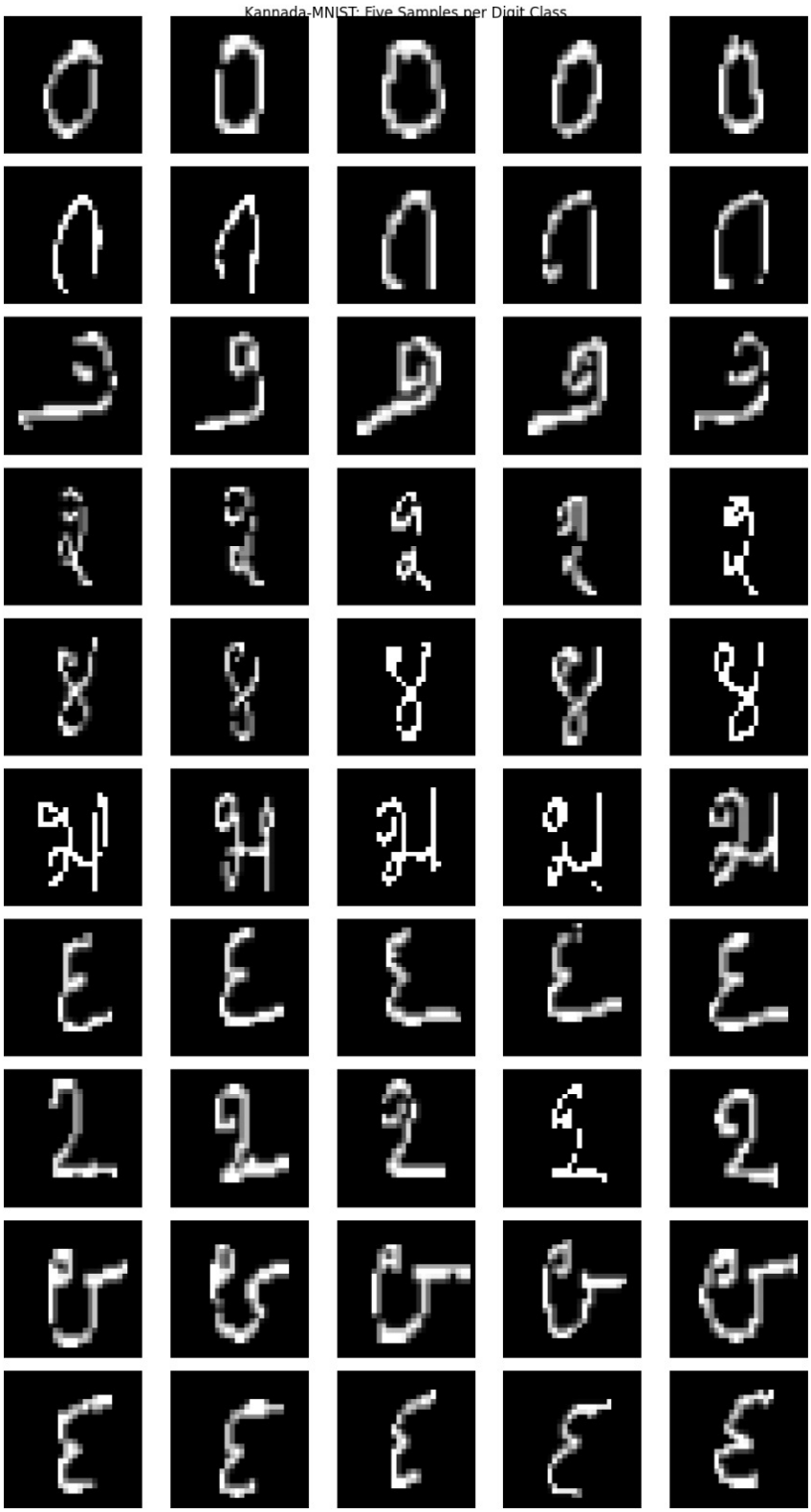


Figure 4. Five handwritten Kannada digit samples for each class (0–9), as supplied with the experimental results.

8. Comparative Observations

• **Validation performance:** LSTM_H128_L2 is best by the assignment's validation-based selection criterion, reaching 99.30%.

• **Held-out evaluation:** GRU_H128_L2 has the highest Dig-MNIST score at 74.36%, narrowly ahead of LSTM_H128_L2 at 74.26%.
• **Training fit:** All models exceed 99.0% training accuracy, and their final training losses are low. High training accuracy does not guarantee equally high performance on Dig-MNIST.
• **Convergence:** Loss drops quickly in early epochs and then stabilizes. RNN_H128_L2 uses all 100 permitted epochs; this is the cap, not evidence that the loss threshold was reached.
• **Gradient flow:** Vanilla RNN has a simpler state update and can be more vulnerable to vanishing/exploding gradients over long sequences. LSTM and GRU gating provides mechanisms that can help preserve useful information. The reported metrics are consistent with strong LSTM/GRU performance, but the experiment does not directly measure gradient norms, so gradient-flow claims remain architectural interpretation rather than a measured result.
• **Class-level errors:** Confusions between classes 3 and 5, 1 and 0, and 4 and 8 are prominent in the supplied matrix. These are candidates for closer visual analysis.

9. Inferences and Conclusion

The experiments demonstrate that row-by-row sequential modeling can classify handwritten Kannada digits effectively on the training/validation split. Treating each image as 28 sequential row vectors makes it possible to apply recurrent architectures to image data without using a conventional convolutional input representation.

Among the tested configurations, LSTM_H128_L2 is the selected model because it has the highest validation accuracy (99.30%). GRU_H128_L2 achieves the highest Dig-MNIST accuracy (74.36%), illustrating that the model with the best validation result is not necessarily the model with the best score on a separate evaluation set. The lower Dig-MNIST accuracies relative to the train/validation scores deserve attention and suggest the need for further testing on representative held-out data.

The training curves show rapid early loss reduction followed by slower improvement. The confusion matrix reveals that overall accuracy alone hides meaningful class-specific errors, particularly between classes 3 and 5, 1 and 0, and 4 and 8. Future experiments could explore bidirectional recurrent layers, additional hidden sizes, learning-rate tuning, regularization, and class-wise error inspection, while keeping the validation split fixed for fair comparison.

10. Limitations and Reproducibility Notes

• The report summarizes Task 1 (Kannada-MNIST) only. Assignment 4 also specifies Task 2, spoken Arabic digit classification from variable-length MFCC sequences; this report does not claim to cover Task 2.

• Dig-MNIST.csv is treated as a labeled additional evaluation set. The unlabeled test(1).csv file cannot be used to compute accuracy unless labels are available.

• Metrics and the confusion matrix were transcribed from the supplied Kannada_MNIST_Results.xlsx workbook and provided figures. The plots should correspond to the same executed notebook run before submission.

• The supplied confusion matrix is for LSTM_H128_L2. It should not be presented as the confusion matrix for GRU_H128_L2, even though GRU_H128_L2 has a slightly higher Dig-MNIST accuracy.

11. Conclusion

Overall, the tested recurrent models learn the Kannada digit training data well. LSTM_H128_L2 is preferred by validation accuracy, while GRU_H128_L2 gives the highest reported Dig-MNIST score. The combination of loss curves, accuracy comparison, confusion matrix, and digit examples provides a more complete evaluation than a single accuracy number. The final interpretation should retain the distinction between validation-based model selection and performance on the additional Dig-MNIST set.

Reference: Assignment 4, ·Sequential Modeling with Recurrent Networks,· LLM course handout, and the Assignment 4 experimental results workbook and figures supplied for this report. The report structure follows the uploaded Assignment 3 Task 1 report.